Agentic Search and the Robustness of Time-Series Foundation Models in Small Open Economies: Evidence from Norway, Canada, and Sweden

Vegard H. Larsen* Leif Anders Thorsrud†

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Abstract

We evaluate whether LLM-guided search can improve pseudo-real-time macroeconomic forecasts from Chronos-2, a time-series foundation model, across three small open economies: Norway, Canada, and Sweden. The search discovers country-specific, economically interpretable covariate pipelines that reduce validation-era MASE by 2.5% (Norway), 3.0% (Canada), and 9.2% (Sweden). However, out-of-sample generalization is heterogeneous. Norway's gains collapse on the 2016–2025 test period (+11.0% degradation), Canada's mostly survive (+3.2%), while Sweden's *improve* further (-1.2%). The key mechanism is that Norway's search relies on LoRA fine-tuning, which overfits; Sweden's uses covariate selection and context truncation, which generalize. Zero-shot Chronos-2 is the most robust single method across all three countries. Chronos-2's predictive bands under-cover uniformly on these macro series—the nominal 80% band covers only 0.695 of realizations—and fine-tuning does not fix this; a lightweight per-series isotonic recalibration closes most of the gap at the 50% level (0.416 to 0.473) and materially narrows it at the 80% level.

Keywords: macroeconomic forecasting, foundation models, automated machine learning, time series, small open economies

JEL codes: C53, C55, E37

*BI Norwegian Business School. Email: vegard.h.larsen@bi.no

†BI Norwegian Business School. Email: leif.a.thorsrud@bi.no

1 Introduction

Time-series foundation models—large neural networks pretrained on billions of observations from diverse domains—have rapidly entered the forecaster's toolkit. Models such as Chronos (Ansari et al., 2024, 2025), TimesFM (Das et al., 2024), and MOIRAI (Woo et al., 2024) promise accurate zero-shot forecasts without task-specific training. Whether they can match or exceed classical methods in rigorous macroeconomic applications is an empirical question with no foregone conclusion.

A related question is whether *automated pipeline search* can add value beyond zero-shot inference. Inspired by Karpathy (2026), we design an outer loop in which a large language model proposes pipeline configurations—covariates, transformations, context window, and fine-tuning settings—evaluated against a rolling validation metric. The agent may select from 14 macroeconomic covariates and adjust fine-tuning hyperparameters, but cannot modify the model architecture.

We investigate these questions across three small open economies: Norway, Canada, and Sweden. Most economies are small and open, yet the empirical forecasting literature disproportionately evaluates methods on US data (Stock and Watson, 2002b). Our three countries share important structural features—independent monetary policy, commodity exposure, high-quality statistical offices—but differ in transmission channels. Norway is an oil-exporting economy; Canada is a large commodity exporter closely tied to the US; Sweden has a diversified manufacturing base oriented toward Europe. Because these economies are unlikely to appear prominently in TSFM pretraining corpora, the zero-shot exercise is a cleaner test of out-of-distribution generalization.

We use a harmonized design across all three countries: four target variables (CPI inflation, industrial production, retail sales, and unemployment) at horizons of 1, 3, 6, and 12 months, evaluated on 120 monthly validation origins (2006–2015) and 111 test origins (2016–2025). Five search comparators—informed LLM, domain-blind LLM, random, greedy stepwise, and a pre-locked manual economist benchmark—operate over the same action space to isolate the value of LLM reasoning and domain knowledge.

Our main findings are fourfold. First, zero-shot Chronos-2 is competitive with strong classical baselines (ARIMA, BVAR) across all three countries, with test-era MASE of 0.996 (Norway), 0.888 (Canada), and 0.938 (Sweden). Second, agentic search discovers country-specific, economically interpretable pipelines—monetary policy and exchange rates for Norway, partner-area GDP for Canada, domestic credit and housing for Sweden. Third, the generalization of these gains is *heterogeneous*: Norway's gains collapse on the test period (+11.0% degradation), Canada's mostly survive (+3.2%), while Sweden's actually *improve* out of sample (-1.2%). Fourth, the mechanism behind this heterogeneity is that Norway's search relies on LoRA fine-tuning, which overfits to validation-era covariate relationships; Sweden's search uses covariate selection and context truncation

without fine-tuning, which proves robust to regime change.

This pattern is itself the paper's central contribution. It demonstrates that agentic pipeline search can add value for macro forecasting, but only when the search discovers representation changes (which covariates to include, how much context to use) rather than parameter changes (fine-tuning weights). Fixed-window validation in non-stationary environments risks overfitting, and the risk is concentrated in fine-tuning—not in covariate selection. For practitioners, the implication is clear: prefer covariate and representation search over fine-tuning when regime changes are likely.

2 Related literature

This paper connects three literatures: macro forecasting, time-series foundation models, and agentic AutoML.

2.1 Macroeconomic forecasting

Data-rich macro forecasting has progressed from dynamic factor models (Stock and Watson, 2002b,a; Forni et al., 2005) through Bayesian VARs with shrinkage priors (Bańbura et al., 2010; Koop and Korobilis, 2013) to machine learning methods that capture nonlinearities (Goulet Coulombe et al., 2022; Babii et al., 2024). Real-time evaluation must contend with the ragged-edge problem of asynchronous publication (Ghysels et al., 2004; Giannone et al., 2008); our pseudo-real-time protocol builds directly on this tradition, using expanding windows and Diebold and Mariano (2002) tests as standard (Clark and McCracken, 2013).

A consistent finding in the macro forecasting literature is that simple models are hard to beat, especially during periods of structural instability. Forecasting small open economies poses particular challenges: oil shocks, exchange rate volatility, and trade-channel disruptions can alter covariate relationships rapidly (Bjørnland et al., 2019; Aastveit et al., 2023). Our cross-country design tests whether automated covariate discovery can adapt to these country-specific transmission channels.

2.2 Foundation models for time series

Time-series foundation models (TSFMs) are pretrained on massive, heterogeneous corpora to enable zero-shot inference on unseen series. Chronos (Ansari et al., 2024) treats forecasting as language modeling via value quantization; its successor Chronos-2 (Ansari et al., 2025) adds group attention that alternates between within-series and cross-series patterns, enabling in-context learning from covariates without architectural changes. Alternative TSFMs include TimesFM (Das et al., 2024, 2025), Lag-Llama (Rasul et al.,

2023), and MOIRAI (Woo et al., 2024; Liu et al., 2025); concerns about pretraining data contamination are documented by Aksu et al. (2024).

The tension between zero-shot inference and task-specific fine-tuning is central to applied TSFM work (Rasul et al., 2023). Our research interrogates this tradeoff by decomposing forecast gains into the foundation model itself, economy-specific adaptation, and the additional value of agent-guided search—across three countries that each present different adaptation challenges.

Separate from point accuracy, the calibration of predictive bands emitted by TSFMs has received much less systematic attention than in classical and ML forecasting, where conformal prediction (Romano et al., 2019; Angelopoulos and Bates, 2023) and isotonic recalibration (Kuleshov et al., 2018) are standard post-hoc remedies. To our knowledge, no prior work has documented TSFM band calibration on small-open-economy macroeconomic series, nor evaluated per-series conformal-style recalibration in that setting; Section 5.8 takes a first step.

2.3 Automated machine learning and feature engineering

Traditional AutoML frameworks (Feurer et al., 2015; Erickson et al., 2020; Shchur et al., 2023) employ Bayesian optimization and meta-learning but operate within rigidly defined search spaces and cannot incorporate domain knowledge. A newer paradigm uses LLMs as meta-optimizers that reason about results and autonomously modify experimental configurations (Lu et al., 2024; Yamada et al., 2025; Cheng et al., 2026; Dong et al., 2025).

Our approach builds on Karpathy's "autoresearch" framework (Karpathy, 2026), which uses a ratchet loop with fixed data preparation, a mutable training script, and a natural-language specification of the objective. This has been adopted across disciplines (O'Connell, 2026; Weidener et al., 2026; Liu et al., 2026) but not yet in macroeconomic forecasting. We adapt it by constraining the search to data representations, covariate selection, and fine-tuning—never model architecture—with pseudo-real-time evaluation governed by rolling forecast-origin validation loss.

3 Methodology

3.1 Foundation model: Chronos-2

We use Chronos-2 (Ansari et al., 2025) as the foundation model backbone, accessed via the AutoGluon TimeSeries interface (Shchur et al., 2023). The model has 120 million parameters and supports a maximum context length of 8,192 time steps. Its group attention mechanism alternates between time attention (within a single series) and group

attention (across series in a batch), enabling in-context learning from covariates without requiring explicit covariate-specific input channels.

Chronos-2 natively integrates both past covariates and known future covariates during inference. In our implementation, all macroeconomic covariates are provided as *past covariates only*: the model receives historical values up to the forecast origin but no future values. This is critical for pseudo-real-time validity.

Fine-tuning. Chronos-2 supports parameter-efficient adaptation via Low-Rank Adaptation (LoRA), targeting the self-attention Q, K, V, O projections and the output patch embedding with rank $r = 8$ and $\alpha = 16$. Fine-tuning is performed once on the data available at the first forecast origin, and the adapted model is reused for all subsequent origins. This fit-once architecture means the fine-tuned model may overfit to early-period patterns—a limitation we return to in Section 6.

3.2 Agentic search procedure

The search is orchestrated by an outer loop that uses a large language model (Claude Sonnet, via the Anthropic API) to propose pipeline configurations. The loop follows a ratchet design inspired by Karpathy (2026):

Algorithm 1 LLM-guided pipeline search

```
Establish baseline: evaluate zero-shot Chronos-2 on all 120 validation origins
for i = 1, 2, ..., N do
    LLM reads: task description, search space, current best config, iteration history
    LLM proposes a JSON configuration (covariates, transforms, context length, fine-tuning)
    Quick evaluation: run on 20 subsampled origins
    if quick score < best quick score so far then
        Full evaluation: run on all 120 origins
        if full score < current best then
            Accept: update best configuration and score
        else
            Reject
        end if
    else
        Reject
    end if
    Log iteration (config, scores, status) to experiment log
end for
```

The two-phase evaluation balances iteration speed with evaluation fidelity. The primary metric is the mean absolute scaled error (MASE), averaged across all target variables and all four forecast horizons.

Prompt design. The search is run independently for each country. In the *informed* variant, the LLM receives a country-specific prompt containing domain knowledge (e.g., "Norway is an oil-exporting economy; Brent crude and exchange rates are highly relevant"). In the *domain-blind* variant, the prompt is stripped of all country-specific hints—variable names are generic. This allows us to isolate the value of domain knowledge in the search.

Search comparators. We compare five search strategies operating over the same action space: (i) informed LLM search, (ii) domain-blind LLM search, (iii) random search (uniform sampling), (iv) greedy stepwise search (deterministic, non-LLM), and (v) a pre-locked manual economist benchmark (covariates chosen by the authors before seeing results).

3.3 Search space

Table 1 summarizes the search dimensions available to the agent.

Table 1: Search space dimensions

Dimension	Range	Default
Covariates	Any subset of 14 variables	None
Covariate transforms	none, log_diff, pct_change, standardize, MA	None
Context length	24, 36, 48, 64, 96, 128, unlimited	Unlimited
Fine-tuning	Off / LoRA	Off
Fine-tune steps	100, 500, 1000, 2000	1000
Fine-tune learning rate	10^{-6} to 10^{-4} (log scale)	10^{-5}
Grouping	Univariate / All targets	Univariate

Target variables are never transformed, as forecasts must remain in the original scale for evaluation. The 14-covariate pool follows a harmonized template across countries (see Section 4.1); exact series differ by country but the categories are aligned.

4 Data and evaluation protocol

4.1 Three-country macroeconomic panels

We construct harmonized monthly panels for Norway, Canada, and Sweden, each following a common template of four target variables and 14 candidate covariates. The four targets are CPI inflation (12-month rate of change), industrial production (seasonally adjusted), retail sales (volume index, seasonally adjusted), and the unemployment rate (LFS, seasonally adjusted).[1]

The 14 covariates span domestic indicators (house prices, credit, exports, imports, the policy rate, bilateral exchange rates against EUR and USD) and global variables (Brent crude, a broad equity index, the federal funds rate, US CPI, VIX, global economic policy uncertainty, and a partner-area activity variable). The partner-area variable is euro-area GDP for Norway and Sweden, and US GDP for Canada, reflecting different trade orientations.

Data sources are country-specific: Statistics Norway (SSB) and Norges Bank for Norway; Statistics Canada and the Bank of Canada for Canada; Statistics Sweden (SCB) and the Riksbank for Sweden. Global variables are sourced from FRED for all three countries. The full variable catalog, including series identifiers and publication lags, is provided in the project repository.

4.2 Pseudo-real-time discipline

At each forecast origin t, we restrict the information set to data that would have been publicly available on date t. For a monthly observation indexed at month-end m, the observation is available when

$$\text{month_end}(m) + \text{publication_lag} \leq t. \tag{1}$$

Publication lags are configured per country and per variable based on official release schedules. Lags range from 0 days (policy rates, known in real time) to 90 days (euro-area GDP). Daily series are aggregated to monthly averages; quarterly series are forward-filled. We use latest-vintage data throughout but restrict availability by publication lag. This is pseudo-real-time, not true real-time with data vintages.

[1] Sweden's retail sales series from Statistics Sweden begins only in 2023 and is excluded from the analysis. Sweden results are computed over three targets (CPI, industrial production, unemployment). See the online appendix for details.

4.3 Evaluation design

We use a rolling expanding-window design. For each forecast origin t, the model receives all data available up to t and produces point forecasts at horizons $h \in \{1, 3, 6, 12\}$ months ahead for each target variable.

Eras. The validation era spans January 2006 to December 2015 (120 monthly origins), used for search and model selection. The test era spans January 2016 to March 2025 (111 origins), evaluated with the pipeline frozen from the search phase. The evaluation window is common across all three countries. We further decompose the test era into three subperiods: pre-COVID (2016–2019), COVID (2020–2021), and post-COVID/inflation (2022–2025).

Metrics. The mean absolute scaled error (MASE) is defined as $\text{MASE}_h = \text{MAE}_h / \text{MAE}_h^{\text{RW}}$, where MAE_h^{RW} is the MAE of the h-step-ahead random walk forecast. The random walk scores exactly 1.000 by construction. MASE is the primary metric because it is scale-independent across targets with different units.

4.4 Baselines

We evaluate a comprehensive set of baselines:

Univariate baselines.

1. **Random walk:** $\hat{y}_{t+h|t} = y_t$.
2. **Seasonal naive:** $\hat{y}_{t+h|t} = y_{t+h-12}$.
3. **AR(p):** Direct autoregressive regression, lag order selected by BIC over $p \in \{1, \ldots, 12\}$.
4. **ARIMA:** Order selected by AIC over $p \in \{0, 1, 2, 3\}$, $d \in \{0, 1\}$, $q \in \{0, 1, 2\}$.
5. **ETS:** Exponential smoothing with automatic model selection by AIC.

Multivariate baselines.

6. **VAR:** Vector autoregression on the target plus the top covariates from the respective country's search. Lag order selected by BIC.
7. **Factor model:** Principal components from all 14 covariates used as regressors in direct h-step forecasting equations.
8. **BVAR:** Minnesota-shrinkage Bayesian VAR, tuned on validation data.
9. **Elastic net:** Direct forecasting with lagged target and lagged covariates, penalty tuned by cross-validation.

Foundation model methods.

10. **Chronos-2 (zero-shot):** No covariates, no fine-tuning.
11. **Chronos-2 (agent-tuned):** Best configuration from the informed LLM search.
12. **Chronos-2 (manual):** Pre-locked economist benchmark—covariates chosen by the authors before seeing results, not revised after.

All baselines are re-estimated at each forecast origin using only data available at that origin, ensuring full pseudo-real-time compliance.

Statistical inference. We test for equal predictive ability using the Diebold and Mariano (2002) test with Newey-West HAC standard errors (bandwidth $h - 1$).

5 Results

5.1 Baseline forecast performance

Table 2 reports test-era MASE by country and horizon. Zero-shot Chronos-2 is competitive with ARIMA and BVAR across all three countries. The random walk scores 1.000 by construction; methods below 1 beat it.

Table 2: Test era MASE by country and method (average across targets)

Country	Method	h=1	h=3	h=6	h=12
Norway	Random walk	1.000	1.000	1.000	1.000
	ARIMA	**0.977**	0.992	1.032	1.029
	BVAR	1.003	1.031	1.109	1.160
	VAR	0.979	1.075	1.175	1.202
	Factor model	1.192	1.170	1.260	1.377
	Chronos-2 (zero-shot)	0.984	**0.981**	**0.997**	**1.021**
	Chronos-2 (agent-tuned)	1.011	1.025	1.085	1.205
Canada	Random walk	1.000	1.000	1.000	1.000
	ARIMA	1.024	1.006	0.982	0.964
	BVAR	1.071	1.124	1.246	2.433
	VAR	1.007	1.222	1.844	21.635
	Factor model	1.459	1.340	1.161	0.978
	Chronos-2 (zero-shot)	0.978	0.930	0.865	0.779
	Chronos-2 (agent-tuned)	**0.947**	**0.914**	**0.854**	**0.763**
Sweden	Random walk	1.000	1.000	1.000	1.000
	ARIMA	0.945	0.960	0.985	1.002
	BVAR	0.966	0.953	0.949	0.945
	VAR	0.927	0.965	1.016	1.066
	Factor model	1.279	1.191	1.349	1.426
	Chronos-2 (zero-shot)	**0.895**	0.925	0.922	1.010
	Chronos-2 (agent-tuned)	0.929	**0.893**	**0.887**	**0.899**

Norway's test-era MASE for zero-shot Chronos-2 is 0.996, marginally better than the random walk and competitive with ARIMA (1.008). Canada shows stronger performance: the zero-shot model achieves 0.888 against ARIMA's 0.994. For Sweden, zero-shot Chronos-2 achieves 0.938, also outperforming ARIMA (0.973). Classical multivariate methods (VAR, factor model) generally underperform their univariate counterparts, consistent with the known difficulty of improving on simple models in macro forecasting.

5.2 Search comparison

Table 3 compares validation-era MASE across the five search methods and three countries. All methods operate over the same action space and budget (50 iterations).

Table 3: Search comparison: validation-era MASE by country and method

Search method	Norway	Canada	Sweden
Informed LLM	0.975	0.843	**0.913**
Blind LLM	0.980	**0.836**	0.925
Random	0.942	0.847	0.936
Greedy	**0.922**	0.841	0.992
Manual economist	0.979	0.926	0.996
Zero-shot baseline	0.999	0.869	1.006

No single search method dominates across countries. For Norway, greedy stepwise search achieves the lowest MASE; for Canada, the domain-blind LLM is best; for Sweden, the informed LLM achieves the strongest result. The manual economist benchmark is competitive in some cases (e.g., Norway) but substantially weaker in others (e.g., Canada). Domain knowledge helps most clearly for Sweden, where the informed LLM outperforms the blind variant.

5.3 Selected pipelines

Table 4 reports the best configuration from the informed LLM search for each country.

Table 4: Best search configuration per country (informed LLM, seed 42)

Country	Covariates	Transforms
Norway	sp500, policy_rate, fed_funds, nok_usd	—
Canada	policy_rate, euro_area_gdp	euro_area_gdp: log_diff
Sweden	policy_rate, fx_usd, house_prices, credit	fx_usd: log_diff, house_prices: pct_change_

All three configurations are economically interpretable. Norway's search discovers equity, monetary, and exchange rate covariates with LoRA fine-tuning. Canada's search identifies partner-area GDP and monetary policy. Sweden's search selects domestic credit, housing, monetary policy, and exchange rates—notably with context truncation to 48 months and *no fine-tuning*. The monetary policy rate appears in all three countries, reflecting its universal relevance for macro forecasting. Fine-tuning is selected in two of three countries; its absence in Sweden's best config proves consequential for generalization.

5.4 Test-era generalization

Table 5 presents the central result: the validation-to-test generalization gap for zero-shot and agent-tuned configurations.

Table 5: Validation-to-test generalization gap (average MASE across targets and horizons)

Country	Config	Val	Test	Gap (%)	h=12 Test
Norway	Zero-shot	0.999	0.996	-0.3%	1.021
	Agent-tuned	0.975	1.081	+11.0%	1.205
Canada	Zero-shot	0.869	0.888	+2.2%	0.779
	Agent-tuned	0.843	0.869	+3.2%	0.763
Sweden	Zero-shot	1.006	0.938	-6.7%	1.010
	Agent-tuned	0.913	0.902	-1.2%	0.899

Generalization is strikingly heterogeneous. Norway's agent-tuned config improves validation MASE by 2.5% but degrades +11.0% on the test period—severe overfitting. Canada shows mild degradation (+3.2%)—most validation gains survive. Sweden's agent-tuned config improves 9.2% on validation and *further improves* on the test period (-1.2%).

Zero-shot Chronos-2 is robust across all three countries: its validation-to-test gaps are small and sometimes negative (indicating improved test performance), consistent with the implicit regularization provided by pretraining on diverse time series.

Figure 1 visualizes the generalization pattern. Points above the 45-degree line indicate overfitting (test MASE worse than validation).

The mechanism behind this heterogeneity is clear from the pipeline comparison: Norway's search relies on LoRA fine-tuning, which adapts model parameters to validation-era covariate relationships that shift under the regime changes of 2016–2025. Sweden's search uses covariate selection and context truncation without fine-tuning—changes to the model's *input representation* rather than its *parameters*—which proves robust.

5.5 Subperiod analysis

Figure 2 decomposes test-era performance into three subperiods. During the pre-COVID period (2016–2019), the agent-tuned configuration is competitive everywhere: the covariate relationships discovered during 2006–2015 still hold. During the COVID period (2020–2021), all model-based methods struggle; the random walk is resilient. During the post-COVID inflation/tightening period (2022+), the overfitting becomes visible: Norway's agent-tuned MASE spikes while Sweden's holds steady.

5.6 Ablation: decomposing the gains

To identify which components drive each country's gains, we perform leave-one-component-out ablation on each country's best configuration. Table 6 and Figure 3 report the

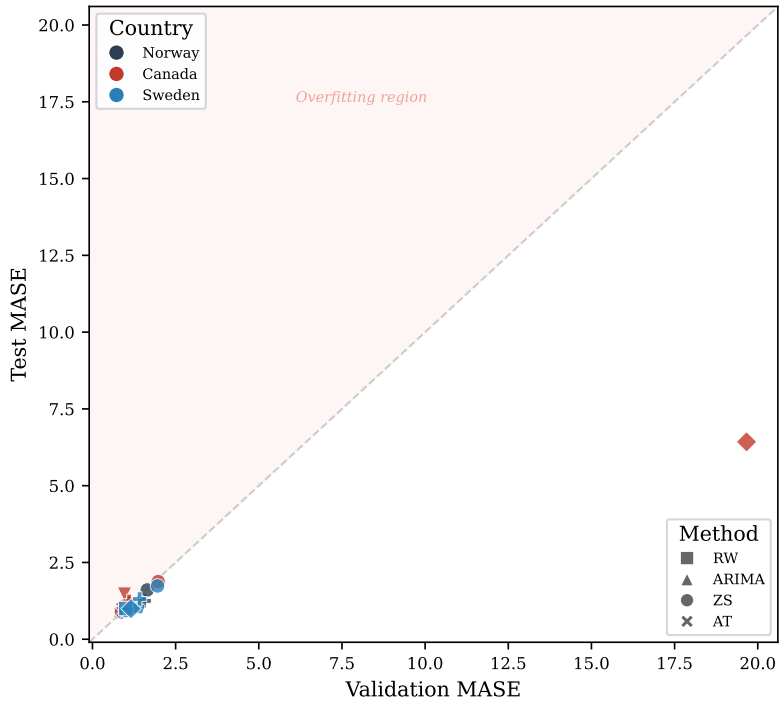

Figure 1: Validation vs. test MASE for all methods and countries. Points above the diagonal indicate overfitting. Norway's agent-tuned config (red X, upper right) shows the largest generalization gap.

validation-era MASE degradation when each component is removed.

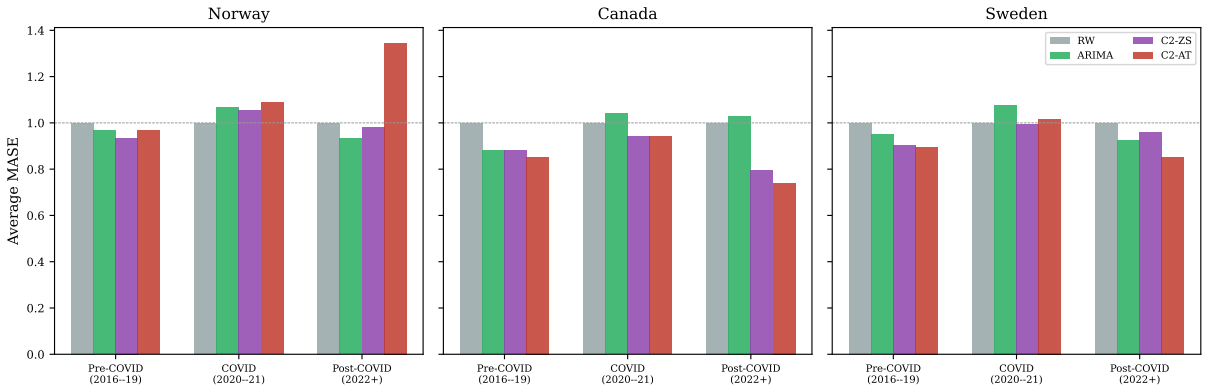

Figure 2: Subperiod MASE by country. The post-2022 inflation/tightening period reveals Norway's overfitting (agent-tuned MASE rises sharply) while Sweden's searched config remains robust.

Table 6: Leave-one-out ablation: MASE degradation (%) when each component is removed

Country	Component removed	MASE	Degrad. (%)
Norway	Full config	0.975	—
	− sp500	0.972	-0.22%
	− policy rate	0.984	+0.97%
	− fed funds	0.990	+1.58%
	− nok usd	0.980	+0.53%
	− fine tune	0.975	+0.07%
	− all covariates	1.003	+2.96%
Canada	Full config	0.843	—
	− policy rate	0.843	+0.00%
	− euro area gdp	0.859	+1.94%
	− fine tune	0.859	+1.97%
	− all covariates	0.859	+1.94%
Sweden	Full config	0.913	—
	− policy rate	0.933	+2.13%
	− fx usd	0.923	+1.04%
	− house prices	0.916	+0.29%
	− credit	0.937	+2.64%
	− context length	0.974	+6.63%
	− all covariates	0.969	+6.09%

The decomposition reveals distinct drivers across countries. For Norway, the federal funds rate is the most valuable covariate (+1.6% degradation when removed), while

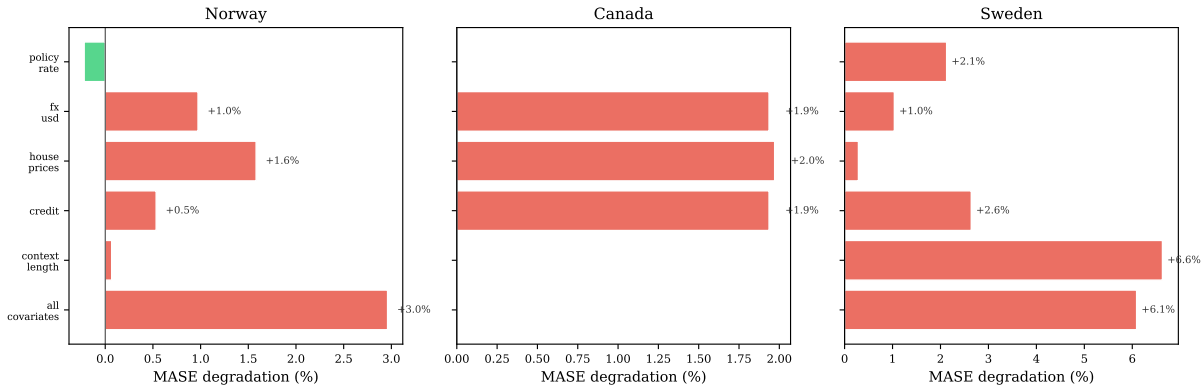

Figure 3: Leave-one-out ablation by country. Bars show MASE degradation (%) when each component is removed from the best configuration. Sweden's context truncation is the single most impactful component across all countries.

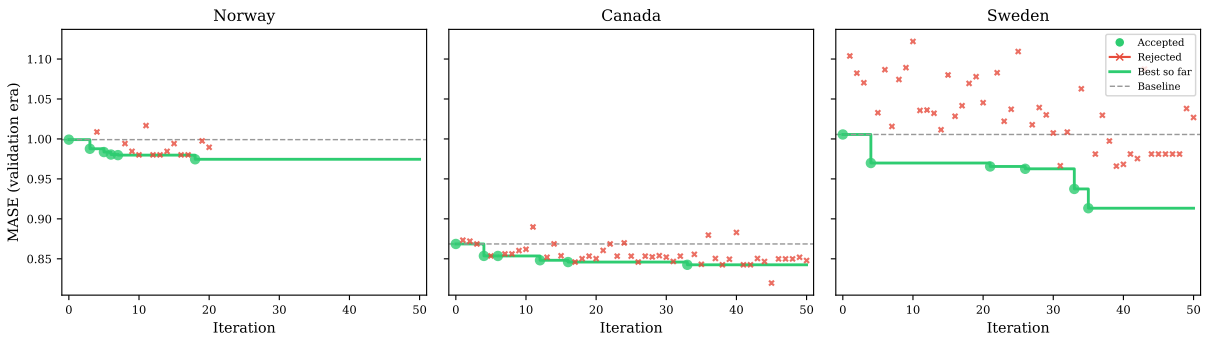

Figure 4: LLM-guided search trajectory by country (50 iterations each). Green circles: accepted improvements. Red crosses: rejected proposals. Step lines show the best-so-far MASE frontier.

fine-tuning contributes almost nothing (+0.1%). For Canada, euro-area GDP drives the entire improvement (+1.9%); the policy rate is dead weight (0.0%). For Sweden, context truncation to 48 months is the single biggest factor (+6.6%), followed by credit (+2.6%) and the policy rate (+2.1%).

A cross-country pattern emerges: *fine-tuning contributes little to validation gains anywhere*, while covariate selection and context window are the high-value levers. This explains the generalization heterogeneity: Norway's "gain" came partly from fine-tuning (fragile under regime change), while Sweden's came entirely from representation changes (robust).

5.7 Search trajectory

Figure 4 shows the informed LLM search trajectory for each country. Norway's search converges by approximately iteration 30, with incremental covariate additions. Canada's search finds fewer improvements (sparser acceptance). Sweden's search finds the largest improvement through a combination of covariates and context truncation.

5.8 Predictive band calibration

The results so far evaluate point forecasts under MASE. Practitioners using macroeconomic forecasts, however, care about predictive bands as much as point estimates: a central-bank projection without a fan chart is of limited operational use. Chronos-2 returns a discrete predictive distribution at each origin as a set of five quantiles $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$, which map directly to nominal 80% (q10–q90) and 50% (q25–q75) prediction intervals. Whether these bands are empirically calibrated—that is, whether 80% of realized outcomes actually fall inside the nominal 80% band—is a separate question from whether the median is accurate. We devote this subsection to that question, because an otherwise accurate point forecast with miscalibrated bands can still mislead downstream risk assessments.

Methodology. We measure empirical coverage for each nominal level $\alpha \in \{0.5, 0.8\}$, pooled across all 3 countries, target variables, forecast horizons $h \in \{1, 3, 6, 12\}$, and 111 test-era origins. For the recalibration exercise, we follow a quantile-based variant of the isotonic recalibration scheme of Kuleshov et al. (2018) that is equivalent in spirit to split-conformal quantile regression (Romano et al., 2019) applied per series. Concretely, for each (country, target, horizon) triple we compute the probability-integral-transform (PIT) values on the *validation* era and take the empirical CDF $\hat{F}$ of those PIT values. At test time, the calibrated quantile forecast at level τ is $Q_{\text{cal}}(\tau) = Q_{\text{base}}(\hat{F}^{-1}(\tau))$, where Q_{base} is the linearly interpolated base quantile function produced by Chronos-2 from its five emitted quantiles. This procedure is local to each series, requires no retraining of the foundation model, and uses validation-era residuals only—fully respecting the pseudo-real-time discipline. It targets marginal, not conditional, calibration.

Universal under-coverage. Table 7 reports per-country, per-target coverage for the 80% and 50% bands, for the fine-tuned agent-selected configuration ("FT"), the zero-shot baseline ("ZS"), and the recalibrated fine-tuned output ("Cal"). The under-coverage is uniform: pooled across the full test set, the fine-tuned 80% band covers only 0.695 of realizations against a nominal 0.80 (a gap of -10.5 pp), and the 50% band covers 0.416 against nominal 0.50 (a gap of -8.4 pp). The zero-shot bands are marginally tighter still, at 0.687 and 0.409 respectively (gaps -11.3 and -9.1 pp). Under-coverage is therefore not an artifact of LoRA adaptation—the raw pretrained Chronos-2 predictive distribution is itself variance-compressed on these macro series, and fine-tuning neither fixes nor meaningfully worsens the problem (less than one percentage point of difference in either band).

Table 7: Empirical coverage of nominal 80% (q10–q90) and 50% (q25–q75) predictive bands. FT: fine-tuned agent-selected Chronos-2. ZS: zero-shot Chronos-2. Cal: per-series isotonic recalibration of the fine-tuned output using validation-era PIT. Pooled rows average across country–target–horizon–origin.

		80% band (q10–q90)			50% band (q25–q75)		
Country	Target	FT	ZS	Cal	FT	ZS	Cal
Norway	CPI	0.682	0.690	0.675	**0.350**	0.357	0.354
	Industrial production	0.775	0.757	0.877	0.479	0.470	**0.670**
	Retail sales	0.722	0.711	0.801	0.430	0.420	0.525
	Unemployment	0.745	0.759	0.773	0.452	0.459	0.506
Canada	CPI	0.727	0.725	0.700	0.504	0.509	0.484
	Industrial production	0.709	**0.638**	0.725	0.455	0.385	0.515
	Retail sales	**0.536**	**0.521**	**0.608**	**0.318**	**0.300**	0.355
	Unemployment	**0.647**	0.662	**0.563**	**0.333**	**0.349**	**0.334**
Sweden	CPI	**0.642**	**0.642**	0.742	**0.346**	**0.346**	0.443
	Industrial production	0.672	0.672	0.763	0.411	0.411	0.517
	Unemployment	0.784	0.784	0.749	0.494	0.494	0.503
Pooled (country×target×horizon)		0.695	0.687	0.725	0.416	0.409	0.473

Directional biases are separable from compression. Pooled coverage gaps conceal a richer story that emerges only when the PIT distribution is examined per series. Table 8 reports the fraction of realizations above the q90 quantile and below the q10 quantile for each (country, target) pair under zero-shot Chronos-2; under perfect calibration each fraction should equal 0.10. Figure 6 plots the full PIT histograms, with the two outer tail bins highlighted. Three qualitatively distinct patterns appear. First, some series are *right-tail heavy*: Canada retail sales places 0.422 of actuals above q90 against a nominal 0.10, and Sweden CPI and industrial production exhibit similar (though milder) right-tail excess, consistent with Chronos-2 underestimating the upside of the 2021–2024 inflation and output-rebound episodes. Second, some series are *left-tail heavy*: Canada unemployment has 0.228 of realizations below q10, reflecting a steep 2020–2021 decline in the unemployment rate that the model did not anticipate. Third, some series show essentially symmetric compression: Norway CPI has close to equal mass in both tails, indicating that the base model is simply too confident rather than systematically biased. This decomposition—compression versus directional bias—matters because the two pathologies call for different remedies. Isotonic recalibration, which is monotone in τ, can correct both compression and bias that is well-approximated by a constant quantile-level shift in the validation era, but cannot correct conditional biases that depend on covariates or regime.

Table 8: PIT tail fractions under zero-shot Chronos-2. Each cell is the share of test-era realizations above q90 or below q10 within a given (country, target) pair, pooled across horizons and origins. Under calibration both columns should equal 0.10.

Country	Target	Below q10	Within 80% band	Above q90
(expected)		*0.100*	*0.800*	*0.100*
Norway	CPI	**0.167**	**0.690**	0.144
	Industrial production	0.072	0.757	**0.171**
	Retail sales	0.092	**0.711**	**0.197**
	Unemployment	0.143	0.759	0.098
Canada	CPI	0.092	**0.725**	**0.183**
	Industrial production	0.082	**0.638**	**0.280**
	Retail sales	0.058	**0.521**	**0.422**
	Unemployment	**0.228**	**0.662**	0.110
Sweden	CPI	0.101	**0.642**	**0.258**
	Industrial production	**0.047**	**0.672**	**0.281**
	Unemployment	0.060	0.784	**0.156**

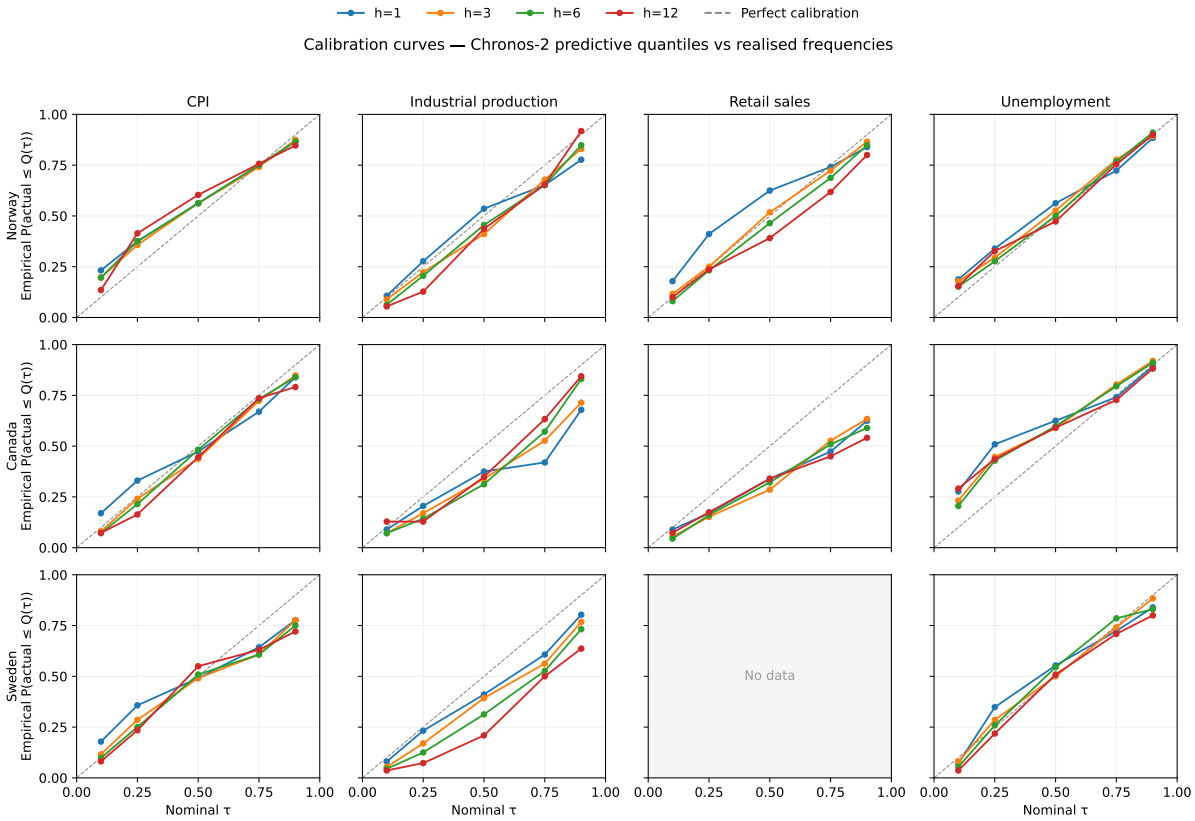

Figure 5: Empirical coverage at nominal quantile levels τ for the fine-tuned Chronos-2 configuration, one panel per (country, target). Coloured lines show the empirical coverage curve $\tau \mapsto \Pr(Y \leq Q(\tau))$ at each horizon; the diagonal is perfect calibration. Curves lying below the diagonal at high τ indicate under-coverage of the upper tail.

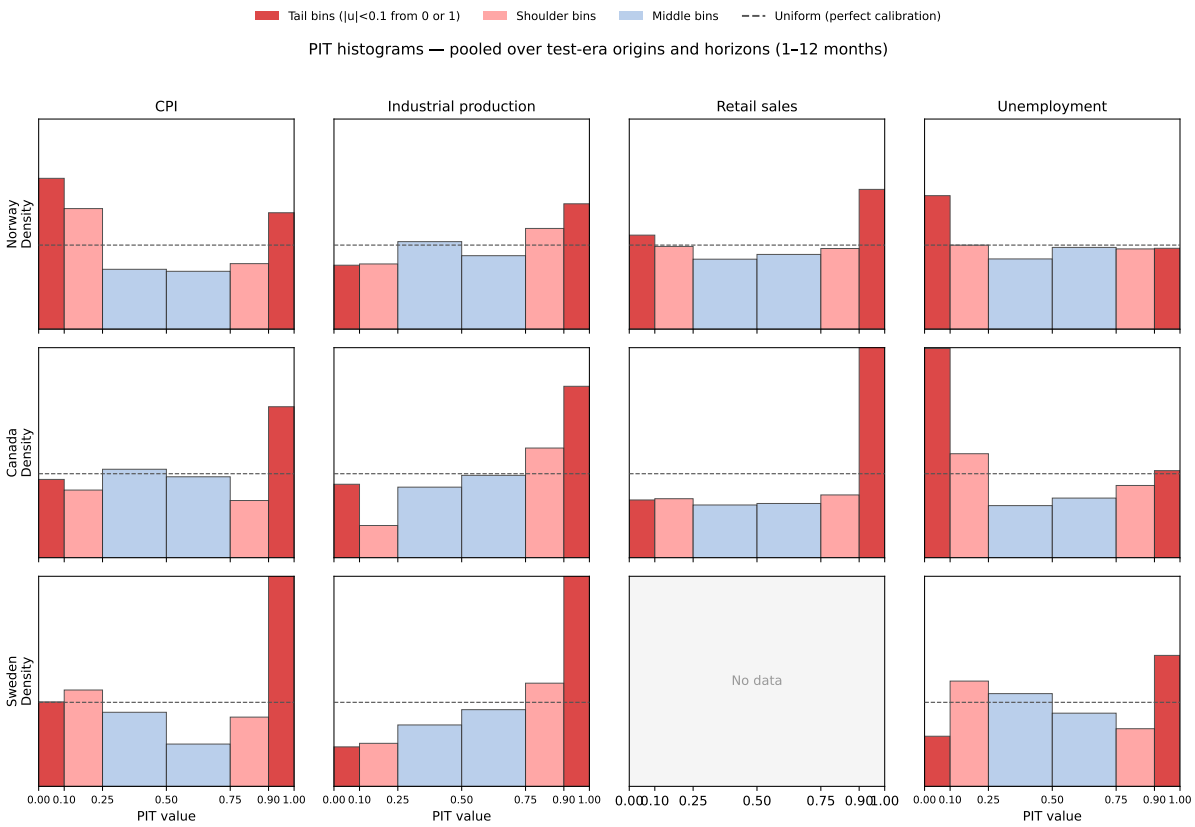

Figure 6: PIT histograms of fine-tuned Chronos-2 on the test era, one panel per (country, target). The two outermost bins (red) are the right-tail and left-tail excess-mass indicators. A flat histogram would indicate perfect calibration; U-shaped histograms indicate variance compression; asymmetric U-shapes indicate directional bias of the kind reported in Table 8.

Recalibration results. The "Cal" column of Table 7 shows the effect of per-series isotonic recalibration. Pooled empirical coverage of the 80% band moves from 0.695 to 0.725, narrowing the nominal–empirical gap from -10.5 to -7.5 pp (about 28% of the gap closed). The 50% band improves more sharply: from 0.416 to 0.473, narrowing the gap from -8.4 to -2.7 pp (about 68% closed). The asymmetry between the two bands is structural. Because the base forecast emits only five quantiles, $\hat{F}^{-1}(\tau)$ saturates at $\tau = 0.1$ and $\tau = 0.9$—there is no base-model support beyond those endpoints for the calibrator to stretch into. Recalibration therefore cannot push an 80% band's coverage all the way to 0.80 when the base distribution already exhausts its tail reach; it can, however, push the 50% band close to nominal because both 0.25 and 0.75 remain interior to the base-quantile grid. A denser base quantile grid would lift this ceiling.

Per-series, recalibration is strongest where the validation-era PIT distribution resembles the test-era PIT distribution, and weakest under regime change. Sweden CPI and Norway retail sales each gain roughly 8–10 pp of 80%-band coverage; these series show similar tail behaviour in 2006–2015 and 2016–2025. Canada unemployment, in contrast,

loses approximately 8 pp of coverage after recalibration, because the validation-era residuals reflect a labour market that tightened smoothly over 2006–2015 while the test era includes the sharp 2020 spike and 2021 recovery. The heterogeneity parallels the point-forecast generalization finding of Section 5.4: any recalibrator fit to one regime carries a regime-shift risk that no method blind to structural breaks can resolve.

Three findings from this exercise recur in the discussion that follows. First, Chronos-2's predictive bands are uniformly too narrow on these macro series, by roughly ten percentage points at the 80% level. Second, because the zero-shot and fine-tuned variants under-cover to within one percentage point of one another, variance compression is a property of the pretrained foundation model, not of our LoRA adaptation—and, symmetrically, fine-tuning cannot be relied upon to fix band calibration. Third, a lightweight per-series isotonic post-hoc recalibrator closes most of the 50%-band gap and a non-trivial fraction of the 80%-band gap at essentially no computational cost, provided the validation and deployment regimes are reasonably aligned. Calibrated bands should therefore accompany any point forecast from a TSFM of this class.

6 Discussion

Zero-shot robustness. Zero-shot Chronos-2 is the most robust single method across all three countries. Its test-era MASE is below or near 1.0 everywhere, meaning it matches or beats the random walk—a benchmark that is itself hard to outperform during regime changes. Pretraining on diverse time series provides implicit regularization against overfitting to specific covariate relationships. For practitioners, zero-shot foundation models should be a standard baseline in macroeconomic forecasting.

When does agentic search help? The answer is conditional on mechanism. Covariate selection and context truncation (as in Sweden) generalize because they change the model's *input representation* without altering its parameters. LoRA fine-tuning (as in Norway) adapts model weights to validation-era patterns that may not persist. This is consistent with the machine learning principle that representation learning is more robust than parameter tuning on small datasets—and 120 monthly origins is a small dataset by any standard.

Domain knowledge. Informed LLM search finds more interpretable pipelines than domain-blind search in all countries. But interpretability does not guarantee robustness: Norway's economically interpretable covariates (oil, policy rate, exchange rate) still overfit. The blind agent discovers risk and equity proxies (S&P 500, VIX) that the informed agent misses. For the paper's central question, domain knowledge narrows the search toward macro-relevant covariates but does not protect against validation-era overfitting.

Country heterogeneity. The three countries present different generalization patterns that would be invisible in a single-country study. Had we evaluated only Norway, the conclusion would be that agentic search is counterproductive. Had we evaluated only Sweden, the conclusion would be that agentic search delivers large, robust gains. The cross-country evidence reveals the conditional nature of the result: *what* the search discovers matters more than whether it discovers something.

Limitations. We use a single TSFM (Chronos-2); results may differ for TimesFM or MOIRAI. Our pseudo-real-time protocol uses latest-vintage data, not true real-time with data revisions. Our density evaluation in Section 5.8 uses Chronos-2's five emitted quantiles; a richer quantile grid or a continuous ranked probability score (CRPS) could reveal finer patterns, particularly in the tails where our isotonic recalibrator saturates. Sweden evaluates three targets (retail sales excluded due to a data gap); cross-country MASE averages are not perfectly comparable. The search budget of 50 iterations may not exhaust the search space, though diminishing returns are observed in all countries.

7 Conclusion

We evaluate agentic search over covariates, context window, and fine-tuning settings for Chronos-2 across three small open economies. The search discovers country-specific, economically interpretable pipelines in all three countries. However, out-of-sample generalization is heterogeneous: fine-tuning-based gains collapse under regime change, while representation-based gains survive. Zero-shot Chronos-2 is the most robust single method.

Two practical implications follow. First, zero-shot foundation models should be standard baselines in macro forecasting—their robustness to structural breaks is a concrete advantage. Second, when using automated search, prefer covariate selection and context window adjustments over fine-tuning in non-stationary environments. The cross-country evidence shows that *what* the search discovers is more important than *whether* it discovers something. Future work should explore rolling validation windows, multiple TSFMs, and extension to additional economies.

Beyond point accuracy, we document that Chronos-2's predictive bands are systematically too narrow on these three economies—the pooled 80% band covers only 0.695 of realizations, and the 50% band only 0.416—and that fine-tuning neither causes nor corrects this compression. A per-series isotonic recalibrator fit to validation-era PIT residuals closes roughly 28% of the 80%-band gap and roughly 68% of the 50%-band gap, with the remaining 80%-band shortfall attributable to the base model's finite quantile support. The practical takeaway mirrors the point-forecast message: zero-shot foundation models are robust but under-confident, and the right response is to accompany point forecasts with a lightweight, locally-fit calibration layer rather than to trust raw TSFM

quantile output.

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A Per-variable test results

Table 9 reports per-variable MASE for the test era across key methods, stratified by country. Values below 1 indicate improvement over the random walk.

Table 9: Test era MASE by variable and country

Country	Variable	h	RW	ARIMA	BVAR	C2-ZS	C2-AT
Norway	CPI	1	1.000	1.019	**1.010**	1.024	1.026
		3	1.000	**1.015**	1.043	**1.016**	1.055
		6	1.000	0.996	1.068	**0.992**	1.087
		12	1.000	0.998	**0.954**	0.956	1.062
	Industrial prod.	1	1.000	1.043	**1.007**	1.023	1.115
		3	1.000	1.007	0.999	**0.981**	1.092
		6	1.000	1.067	1.105	**0.974**	1.170
		12	1.000	1.056	1.434	**1.047**	1.435
	Retail sales	1	1.000	**0.971**	0.999	0.993	1.022
		3	1.000	1.003	1.038	**0.977**	1.027
		6	1.000	1.069	1.154	**1.036**	1.108
		12	1.000	**1.068**	1.147	1.102	1.320
	Unemployment	1	1.000	**0.876**	0.998	0.896	0.880
		3	1.000	0.941	1.045	0.950	**0.924**
		6	1.000	0.998	1.110	0.988	**0.977**
		12	1.000	0.994	1.107	**0.978**	1.003
Canada	CPI	1	1.000	1.018	**0.993**	1.072	1.010
		3	1.000	**0.968**	1.014	1.045	0.999
		6	1.000	**0.951**	1.022	1.022	0.984
		12	1.000	0.927	0.972	0.918	**0.838**
	Industrial prod.	1	1.000	0.972	1.125	0.875	**0.788**
		3	1.000	0.941	1.107	0.800	**0.744**
		6	1.000	0.918	1.407	0.707	**0.641**
		12	1.000	0.930	3.074	0.602	**0.573**
	Retail sales	1	1.000	1.027	1.123	0.985	**0.977**
		3	1.000	0.991	1.323	**0.928**	0.948
		6	1.000	0.937	1.456	**0.826**	0.852
		12	1.000	0.907	4.470	**0.685**	0.704
	Unemployment	1	1.000	1.078	1.044	**0.979**	1.015
		3	1.000	1.125	1.055	**0.948**	0.965
		6	1.000	1.120	1.099	**0.905**	0.939
		12	1.000	1.093	1.213	**0.910**	0.934
Sweden	CPI	1	1.000	0.979	0.963	**0.883**	0.939
		3	1.000	0.940	0.946	0.853	**0.832**
		6	1.000	0.957	0.931	0.841	**0.810**
		12	1.000	0.967	0.983	0.845	**0.823**
	Industrial prod.	1	1.000	**0.931**	0.948	0.960	0.946
		3	1.000	0.960	**0.947**	1.016	0.957
		6	1.000	1.005	**0.955**	1.077	0.986
		12	1.000	1.002	**0.980**	1.335	1.042
	Unemployment	1	1.000	0.926	0.986	**0.842**	0.903
		3	1.000	0.979	0.965	0.908	**0.889**

B Subperiod details

Table 10 reports average RMSE across targets by subperiod and country.

Table 10: Subperiod analysis (average RMSE across targets)

Country	Subperiod	Method	h=1	h=3	h=6	h=12
Norway	Pre-COVID (2016–19)	RW	0.966	1.138	1.836	2.654
		ARIMA	0.897	1.118	1.836	**2.618**
		C2-ZS	**0.862**	**1.033**	**1.773**	2.633
		C2-AT	0.882	1.081	1.844	2.739
	COVID (2020–21)	RW	**2.328**	**2.921**	**3.149**	**3.411**
		ARIMA	2.491	3.045	3.609	4.130
		C2-ZS	2.519	3.008	3.495	4.003
		C2-AT	2.466	3.086	3.605	4.234
	Post-COVID (2022+)	RW	0.963	1.207	1.374	1.693
		ARIMA	**0.900**	**1.118**	**1.326**	**1.645**
		C2-ZS	0.953	1.183	1.334	1.734
		C2-AT	1.023	1.324	1.641	3.011
Canada	Pre-COVID (2016–19)	RW	202807.636	326540.027	807441.989	907840.929
		ARIMA	**189581.765**	**298078.036**	798782.884	**828279.598**
		C2-ZS	201119.988	304713.511	**788434.724**	835671.376
		C2-AT	200129.068	301445.299	798417.376	853001.526
	COVID (2020–21)	RW	**1608413.258**	**1644085.369**	1400123.782	1816402.652
		ARIMA	1694005.343	1744582.672	1494186.738	1895543.314
		C2-ZS	1665727.876	1709717.234	**1398224.176**	**1630246.493**
		C2-AT	1714283.082	1775714.956	1468558.477	1703626.520
	Post-COVID (2022+)	RW	379455.813	510907.841	617741.243	951556.219
		ARIMA	382193.917	510797.943	616737.670	951582.516
		C2-ZS	374291.357	**455242.375**	**479297.045**	**592155.823**
		C2-AT	**373589.419**	466801.008	503475.987	615251.037
Sweden	Pre-COVID (2016–19)	RW	0.882	1.013	1.997	2.511
		ARIMA	0.795	0.945	1.956	2.495
		C2-ZS	**0.775**	**0.920**	2.002	2.571
		C2-AT	0.807	0.932	**1.939**	**2.490**
	COVID (2020–21)	RW	**2.924**	3.463	**3.374**	**4.621**
		ARIMA	2.974	3.616	3.824	5.274
		C2-ZS	3.059	3.776	3.865	4.929
		C2-AT	2.972	**3.420**	3.379	4.749
	Post-COVID (2022+)	RW	1.782	2.382	2.907	3.787
		ARIMA	1.615	2.162	2.744	3.681
		C2-ZS	**1.552**	2.168	2.969	4.901
		C2-AT	1.628	**2.123**	**2.713**	**3.486**

C Reproducibility

Software environment

Component	Version
Python	3.11.15
`chronos-forecasting`	2.2.2
`autogluon.timeseries`	1.5.0
PyTorch	2.11.0+cu130
`statsmodels`	0.14.6
`anthropic` (Claude API)	0.86.0

All dependencies are pinned in `uv.lock`. Hardware: 2× NVIDIA GeForce RTX 4090 (24 GB VRAM each).

Foundation model

Chronos-2 is loaded via AutoGluon using the model identifier `amazon/chronos-2` (120M parameters). LoRA fine-tuning targets the self-attention Q, K, V, O projections and the output patch embedding with rank $r = 8$ and $\alpha = 16$.

Search agent

The search agent uses Claude Sonnet via the Anthropic Messages API with `max_tokens=1024`. Country-specific informed prompts are in `prompts/informed_{country}.md`; the domain-blind prompt is `prompts/blind.md`. Each search runs for 50 iterations with the two-phase quick/full evaluation protocol.

Data sources

All data sources are publicly available:

- **Norway:** Statistics Norway (SSB) JSON-stat2 API; Norges Bank SDMX API.
- **Canada:** Statistics Canada CSV API; Bank of Canada Valet API.
- **Sweden:** Statistics Sweden (SCB) PxWeb API; Riksbank API.
- **Global:** FRED via the `fredapi` client (free API key required).
- **LLM search:** Anthropic API (required for search; not required for baselines).

Publication lags are configured per country in `configs/publication_lags.yml`. The variable catalog is in `metadata/variable_catalog.csv`. Manual economist benchmark configurations are in `configs/manual_economist_benchmarks.yaml`.

Replication

```
uv sync --extra ml --extra dev
uv run python src/prepare.py            # Norway
uv run python src/prepare_canada.py     # Canada
uv run python src/prepare_sweden.py     # Sweden
uv run python src/baselines.py --all --era validation --save --country norway
uv run python src/baselines.py --all --era test --save --country norway
# Repeat for canada and sweden
HF_HUB_OFFLINE=1 uv run python src/search.py \
    --country norway --mode llm --seed 42 \
    --program prompts/informed_norway.md --max-iterations 50
# Repeat for canada and sweden
uv run python src/build_forecast_errors.py
uv run python src/tables/generate_tables.py
uv run python scripts/generate_figures.py
uv run pytest  # 131+ tests
```

D Zero-shot vs. fine-tuned calibration

Figures 7 and 8 overlay the calibration curves and PIT histograms for zero-shot and fine-tuned Chronos-2 on the same test-era origins. The two variants are visually indistinguishable on most panels, consistent with the sub-one-percentage-point pooled-coverage difference reported in Table 7.

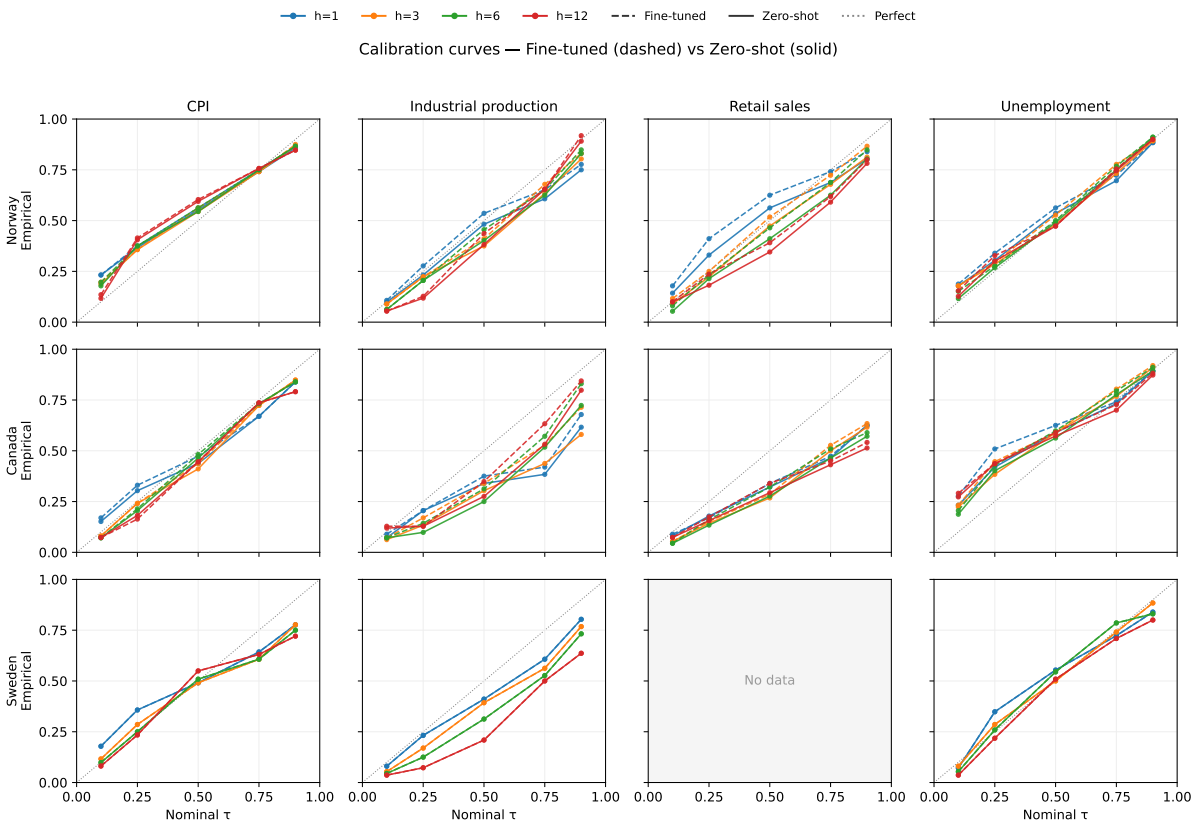

Figure 7: Calibration curves: zero-shot (dashed) versus fine-tuned (solid) Chronos-2, per (country, target) panel.

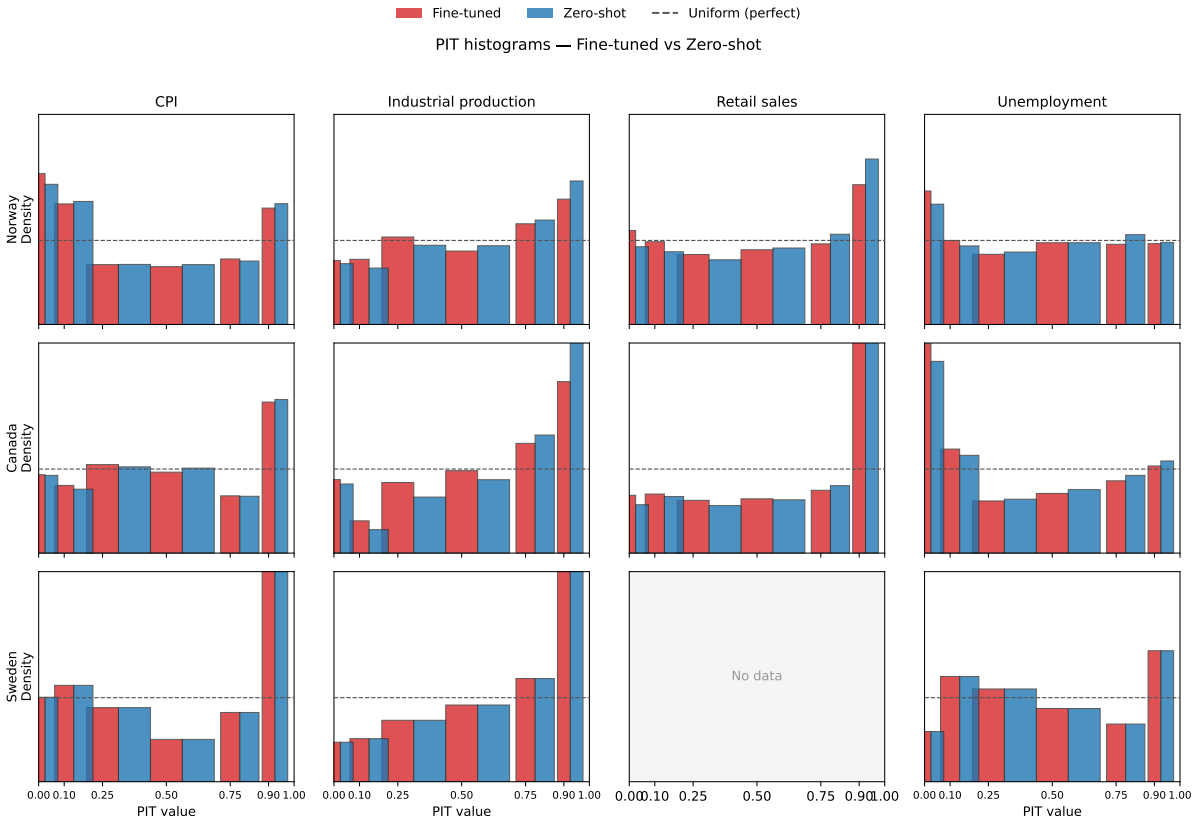

Figure 8: PIT histograms: zero-shot (red) versus fine-tuned (blue) Chronos-2, per (country, target) panel.